

Scaled Dot-Product Attention

$$\text{Attention}(Q, K, V) = \text{softmax}(Q K^T / \sqrt{d_k}) V$$

$X \in \mathbb{R}^{n \times d}$ (input sequence)

W_Q

Q

queries

W_K

K

keys

W_V

V

values

$Q K^T / \sqrt{d_k}$

similarity scores, scaled

softmax(·)

attention weights (rows sum to 1)



×

V

Output $\in \mathbb{R}^{n \times d_k}$

each query's weighted average over the values